

SafeSpeed

Is the posted speed limit survivable? A Safe System screening of the road networks of Maharashtra and Thailand

Methodology report · AI for Safer Roads 2026 Innovation Challenge (Asian Development Bank / Agilysis) · September 2026

Key terms

Term	Meaning	Term	Meaning
v50, v85	median and 85th-percentile traffic speed (km/h)	CLIP, DINOv3	pretrained image models; an embedding is the vector they make from a photo
Safe speed	highest speed at which the crash a road allows is survivable for its most exposed user	VLM	vision-language model: answers questions about an image in text
Ladder, rung	safe speeds by crash type; a rung is one	iRAP, ThaiRAP	International Road Assessment Programme and its Thai arm
Limit gap	posted limit minus safe speed (mismatch)	POI; VRU	point of interest (school, stop, shop ...); vulnerable road user: on foot, cycling or on a motorcycle
Headline	the SafeSpeed score and its safe speed	GHS-SMOD	Global Human Settlement Layer settlement model
KSI	killed or seriously injured: maximum Abbreviated Injury Scale 3+ or fatal (MAIS3+F)	MH, TH; OSM	Maharashtra, Thailand; OpenStreetMap
Expected harm	chance that a crash kills or seriously injures the most exposed user, over the road's speeds	GBM; PCA	gradient-boosted decision trees; principal component analysis
Design speed	the speed a road's layout and surroundings invite, predicted from context alone	Graph neural network (GNN)	a model that also reads the features of connected road pieces (GraphSAGE)
Segment; road piece	one of the 15,554 road sections we score; a shorter TomTom piece of one (46,548 with a valid profile and ≥ 30 probe passes)	Stack	weighted sum of models' predictions; weights non-negative, fitted out of sample
Probe	a TomTom vehicle whose passes measure speed; also a small classifier on an image embedding	R²; κ; ρ; F1	variance explained (1 = perfect); Cohen's kappa (0 = chance agreement); Spearman rank correlation; harmonic mean of precision and recall
Band	how soon to act: Aligned, Low, Moderate, High, Critical	Spatial CV; out-of-fold	testing on 50 km blocks of road the model never saw; predicted by a model that never saw that block
pp; FSI	percentage points; fatal and serious injuries	Block bootstrap	95 % interval over 1,000 resamples of whole 50 km blocks

Running the models

The three models are in one repository on [GitHub](https://github.com/Shishuii/safe_speed.git) (the package): their code, trained weights, sample inputs with expected results, examples and this report. Clone it and everything runs on a laptop CPU with Python 3.9 or later, numpy and pandas, with no GPU or internet connection.

```
git clone https://github.com/Shishuii/safe_speed.git
cd safe_speed
python3 -m venv .venv
source .venv/bin/activate      # Windows: .venv\Scripts\activate
python -m pip install -r requirements.txt
python run_inference.py --check
python run_vision.py --check
python run_ensemble.py --check
```

Command	What it runs	Writes to outputs/
run_inference.py	Design-speed model: invited v50 and v85, each factor group's share, expected harm. Sample: 200 segments from both countries.	predictions.csv
run_vision.py	Vision model, Thailand: image attributes, safe speed, invited speed, harm; --with-dinov3 adds DINOv3 classifier scores. Sample: 3 segments, 7 street images.	vision_predictions.csv, vision_images.csv
run_ensemble.py	Research ensemble: v85 from context, road graph and images; seven members by default, --stack with_dinov3 adds the DINOv3 member, --stack without_images keeps four. Sample: 9 road pieces of 5 segments.	ensemble_predictions.csv

With --check, each run compares its outputs with the expected results in data/ and ends with check: PASSED. Three examples in examples/ use the models from Python. run_inference.py --input my_roads.csv scores your roads (columns: COLUMNS.md). Embedding new street images (run_vision.py --embed-to NEW.npz) needs pip install -r requirements-vision.txt and a one-off CLIP download (about 1.7 GB); --with-dinov3 adds the DINOv3 backbone (about 1.2 GB). Image models and outputs are non-commercial (Section 9).

1. The problem

How badly people are hurt in a crash depends on how fast the vehicles are moving. Limits are usually set by road class and habit, and drivers settle at the speed the road's layout invites. Neither is checked against the human body. On the injury curve we use, the chance that a person on foot struck at 30 km/h is killed or seriously injured is 11 %; at 50 km/h it is 36 %.

The Safe System approach (Tingvall & Haworth 1999; WHO 2017; OECD/ITF 2018) starts from the fact that people make mistakes. Crashes will happen, so the road has to keep their forces within what a body can survive. For speed this becomes a plain test. Where people walk beside or across traffic, cars must be slow enough for a pedestrian to survive being hit. Where two cars can meet head-on, they must be slow enough for the occupants to survive. The survivable speed depends on who can be hit and how.

India and Thailand carry a heavy share of the world's road deaths. WHO estimates 216,618 road deaths in India in 2021 (15.4 per 100,000 people) and 18,218 in Thailand (25.4 per 100,000). Riders of motorcycles and other two- and three-wheelers account for 45 % of reported deaths in India, and pedestrians for another 19 %. In Thailand two- and three-wheeler riders are 51 % of reported deaths, with 46 % recorded as other or unknown, and these vehicles make up 52 % of all registered vehicles. The national maximum urban limit is 70 km/h in India and 80 km/h in Thailand, far above the speed a pedestrian or a motorcyclist survives (WHO 2023).

The challenge gave us the road networks of Maharashtra and Thailand with posted limits and TomTom probe speeds: the median, the 85th percentile and a probe count for each section. Of 69,966 sections, 15,554 are marked valid for analysis in the challenge data; these are the segments we score (4,010 in Maharashtra, 11,544 in Thailand). We added the sources in Table 1.

Neither network comes with crash records, so no model here can learn where crashes happen or how often. We screen for severity instead: if a crash happens on this road at the speeds measured, how likely is death or serious injury?

A road authority needs a list it can act on. For every segment the tool therefore says what speed the road can safely carry, how far the posted limit sits above it, who is exposed, how soon to act and what to do first. Each score splits into named parts, so every result can be traced back to its inputs.

Some things are out of scope. We do not predict crash counts or rates, and we do not compute benefit-cost ratios. Enforcement and driver behaviour are not modelled. The speed models do not transfer between the two countries (R^2 –0.321 from Maharashtra to Thailand, –0.123 the other way), so the design-speed model is fitted separately for each country. Roads outside the two networks are not covered.

Dataset	What it gives	Used for
Challenge road network (TomTom speeds)	network, posted limit, v50, v85, probe count per section	labels, limit gap, traffic volume
TomTom Traffic Stats, per road piece	19 speed percentiles on 46,548 road pieces	speed distribution, measured harm, research labels
Challenge feature files	counts of schools, stops, shops and other POIs; observed helmet use	exposure; model features; helmet use for reporting only
Asian Development Bank Road Safety Performance Indicators	helmet wearing by region, urban or rural	VRU part of the score
WorldPop 2025, 100 m	residents within 300 m of the road	exposure; model features
GHS-SMOD R2023A, 1 km	settlement class, urban centre to rural	safe speed only
OpenStreetMap	lane count, one-way	safe speed (lane count); labels for two image probes
city2graph	graph measures and road graphs	research models
Mapillary	street images and hosted object detections, mainly Thailand	image models; checks on actions
V-RoAst / ThaiRAP	Thai survey images with iRAP codes by ThaiRAP staff	training the image models
Published research	injury curves, iRAP Toolkit, Power Model	safe speed, harm, actions

Table 1. Data sources. Full citations are in the references.

2. How SafeSpeed works

Figure 1 follows one segment through the method, from the data to the recommended action. The vision model and the research ensemble run beside the score and never change it.

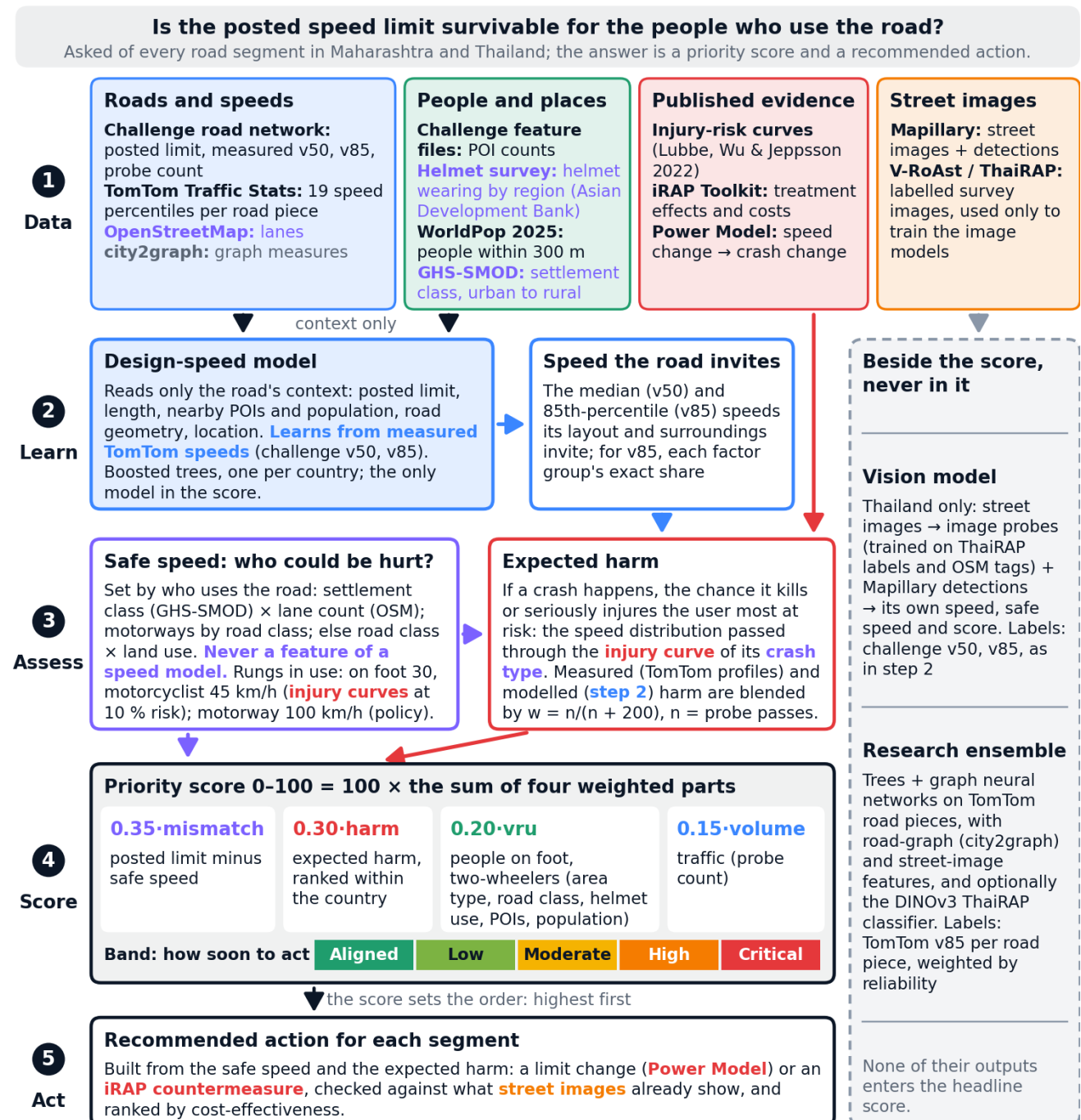


Figure 1. The method in five steps. Purple inputs are never a feature of a speed model; settlement class and lanes set the safe speed. Coloured words tie a step to the data it uses. The models in the dashed panel run beside the score.

3. Labels: measured speed

Our models learn the one measured target we have, speed: log v50 and log v85, the speeds a road's design and surroundings invite. Harm is computed from them afterwards (Figure 2). From TomTom's 19 speed percentiles per road piece we rebuild the full speed distribution (monotone interpolation on the probit scale, Gaussian tails); its mean and standard deviation are off from TomTom's own by 0.14 km/h and 0.56 km/h on average over 1,479 road pieces.

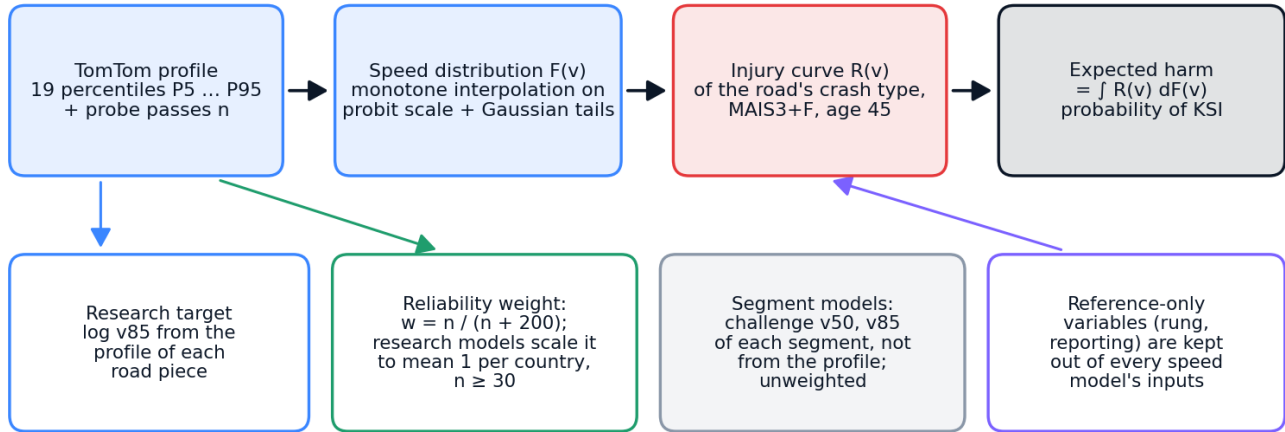


Figure 2. From speed profile to expected harm and labels. The research models learn log v85 from the profile, weighted by reliability; the segment models learn the challenge v50 and v85. The variables that choose the crash type never enter a speed model.

We do not learn harm directly. A harm label is built from the same inputs that choose the crash type (settlement class, lane count), so a model can score well by re-learning that rule. We measure this with a circularity floor, the R^2 reachable from the rule inputs alone, and judge each target by its skill above it. Log v85 has skill 0.336, harm for one crash type 0.158 and harm weighted by the road-user mix -0.244 (Figure 3a). The road-user mix is our estimate of how a segment's crash exposure splits across the crash types of the ladder (Section 5), from land use, road class, nearby POIs, population and the two-wheeler share: the estimated share of motorcycles in traffic, from street images, observed helmet use or the country's share of motorcyclist deaths. Both are used for reporting only. In a wider label study, the label with the highest R^2 ranked fifth on skill (Figure 3b).

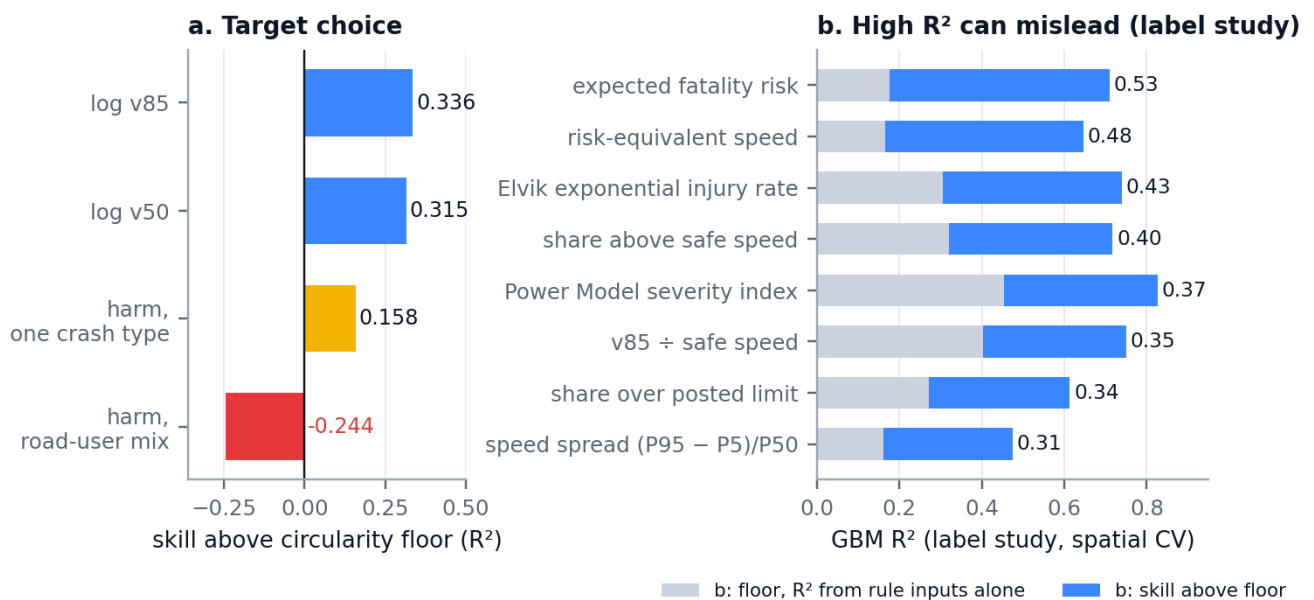


Figure 3. Skill above the circularity floor. a: the target of the design-speed model. b: a separate study of eight candidate labels on the road pieces, each computed over the measured speed distribution: the expected fatality risk; a risk-equivalent speed; Elvik's exponential injury rate and the Power Model's severity index relative to the safe speed; the share of traffic above the safe speed or above the posted limit; v85 ÷ safe speed; and the speed spread. The grey part of each bar is what the rule inputs alone explain. The expected-fatality-risk label takes the crash type from road class and land use only, so it is not the harm target of panel a.

For the same reason, the 22 reference-only columns, which build the safe speed (settlement class, OSM lanes and one-way, the challenge's road class and land use) or describe who uses the road (helmet use, two-wheeler share, road-user mix), never enter a speed model; an automatic check on every speed model's input list enforces this. They would add 0.058 R^2 to the design-speed model. The ban covers the exact columns the safe-speed rules read. Independent measurements of the road's design are allowed, because the target is measured speed: TomTom's functional road class enters the research ensemble, the lane count read from images enters the vision model and the ensemble's image members, and land use read from images by the DINOv3 classifier (Section 4.3) enters only its optional ensemble member.

Neighbouring roads share speeds, so a random split flatters a model. Each speed model is tested on whole 50 km blocks it never saw; a random split would add +0.022 R^2 to the design-speed model (Figure 4).

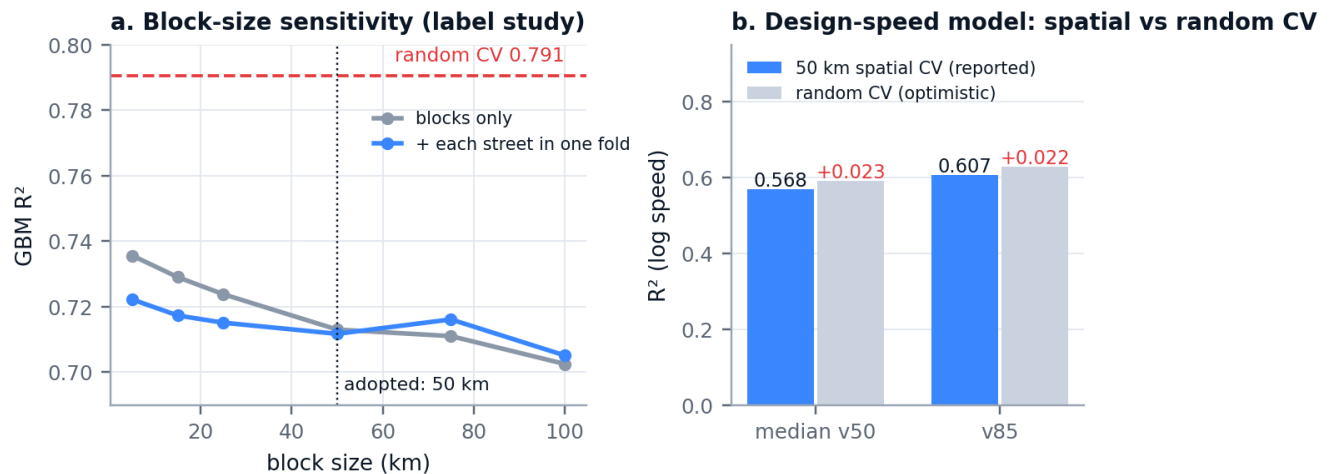


Figure 4. a: R^2 against block size in the label study; grey: blocks only, blue: each named street also kept within one fold (adopted); the dashed line is a random split. b: the design-speed model under spatial and random cross-validation.

4. Models

4.1 Design-speed model

This is the only model inside the score. For each country, and for log v50 and log v85, we fit gradient-boosted trees (depth 2, 1,000 trees, learning rate 0.06) on the 15,143 segments with measured speed. The model reads context only: posted limit, length, nearby schools, transport stops and other POIs, population, road geometry and location. Features may interact only inside six factor groups: location; road geometry (bends and junctions); shops, schools and stops nearby; people living nearby; posted speed limit; and segment length. Each prediction therefore splits exactly into one contribution per group (largest residual 6×10^{-14}). Against a model without this constraint it costs 0.008 R^2 . Figure 5b shows what each design choice adds on the same folds.

Spatial-CV R^2 for log v85 is 0.607 pooled, 0.416 in Maharashtra and 0.512 in Thailand; the mean absolute error is 10.6 km/h. The pooled value also counts the difference between the two countries, so the per-country values are the fairer measure (Figure 5a).

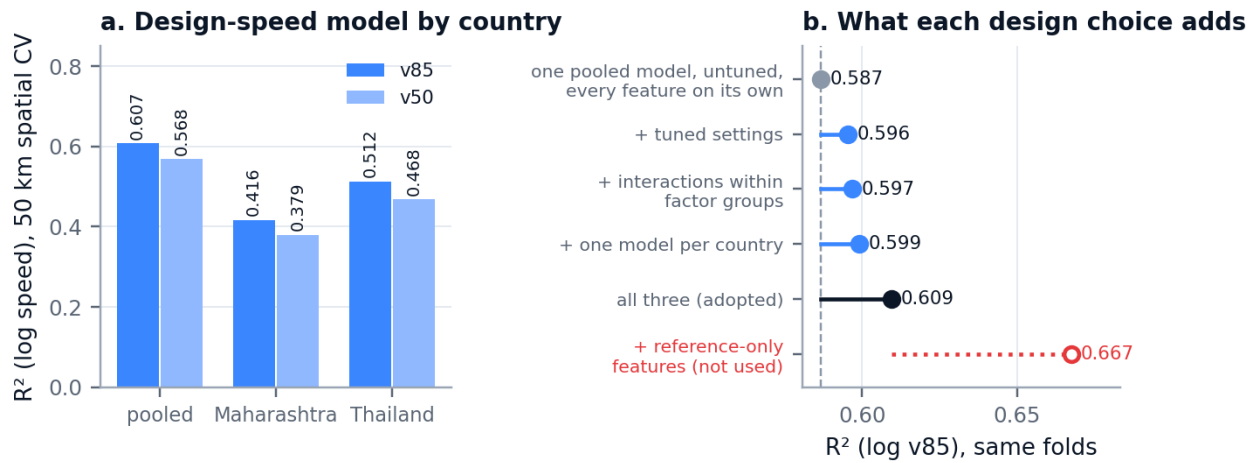


Figure 5. a: R^2 by country and target. b: one ablation run on the same folds. It starts from one pooled model with untuned settings in which every feature acts on its own; each of the next three rows adds one change to it, and the adopted model combines all three. The run ends at 0.609, slightly above the 0.607 of the final model's own cross-validation. Adding the variables that set the safe speed would score higher and is not done.

4.2 Vision model: CLIP probes and Mapillary detections

The vision model asks how much of the speed model can be rebuilt from street images alone. It covers Thailand, where Mapillary imagery is dense. We embed 45,294 images that face along the road with a frozen CLIP ViT-L/14 (Radford et al. 2021). Linear probes trained on ThaiRAP survey labels read 11 iRAP attributes, such as lighting, shoulders, roadside hazards and traffic calming. An attribute is kept only if it reaches $\kappa \geq 0.20$ and beats the majority class on held-out ThaiRAP segments. Two probes trained on OpenStreetMap tags read the lane count and whether the road is divided, and Mapillary's own detections show people, riders, footpaths and crossings. The first 32 principal components of the embedding describe overall appearance (Figure 6).

ThaiRAP labels (V-RoAst) train the probes; each attribute must reach $\kappa \geq 0.20$ and beat the majority class

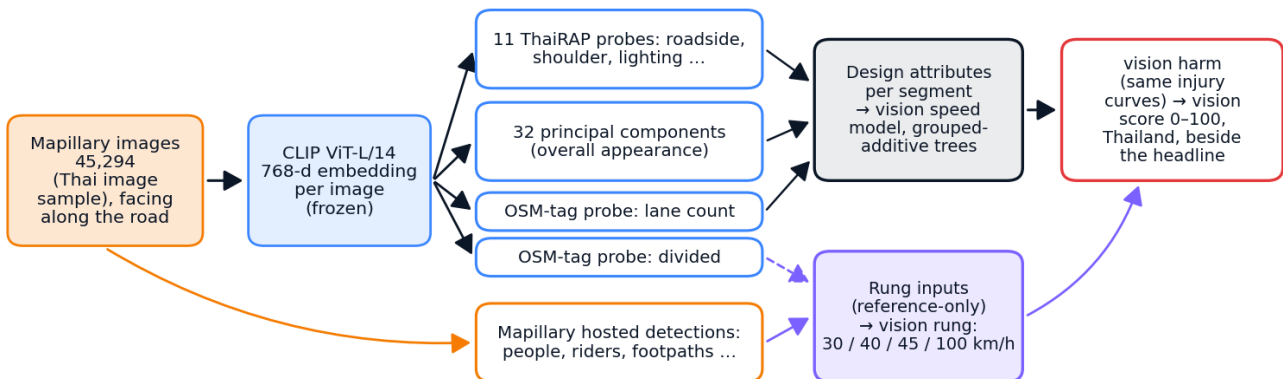


Figure 6. The vision model. Probes and embedding components feed the speed model; detections and the divided-road probe set the vision safe speed.

The vision safe speed uses the same ladder, read from what the images show. A segment gets the pedestrian rung (30 km/h) when people on foot, a footpath or a crossing appear in at least 25 % of its images, the cyclist rung (40 km/h) when cyclists appear in at least 25 %, and otherwise the motorcyclist rung (45 km/h), because motorcycles are assumed on every road. The motorway rung (100 km/h) needs the road to look divided in at least 80 % of the images, no person, rider, footpath, crossing or junction in any of them, and at least 10 images on which the detector ran, because a few images without people are weak evidence that nobody walks there. The vision score applies the score's formula and weights (Section 6) to image inputs: the gap between the posted limit and the vision safe speed, the within-country rank of the vision harm, a VRU part from the shares of images showing people on foot, riders, cyclists, footpaths and crossings, and a volume part from the lane count read from the images, since no image shows traffic.

Of the Thai segments, 6,458 have usable images. On the 6,268 of them with measured speed, the images alone reach v85 R^2 0.437, against 0.523 for the context model on the same rows and 0.569 for both together (Figure 7a). Over all 6,458, the vision safe speed agrees with the headline one on 61 % of segments (Figure 7b). Where they differ, the segment is a candidate for a field check; neither is ground truth.

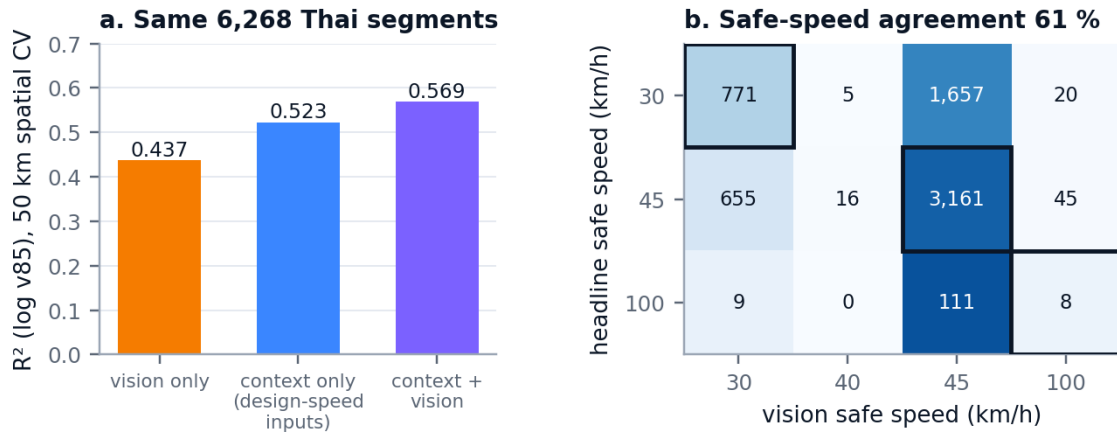


Figure 7. a: images, context and both, on the same 6,268 Thai segments with measured speed. b: headline against vision safe speed on all 6,458 segments with images (segment counts); outlined cells agree.

We also tried coding iRAP attributes by asking vision-language models (VLMs) directly, the method of V-RoAst (Jongwiriyanurak et al. 2025). On ThaiRAP's labels it fails: mean κ 0.11 for Gemini-1.5-flash, 0.05 for GPT-4o-mini and 0.01 for a local Qwen2.5-VL-3B (Figure 8a). On the attributes kept, the CLIP probes reach κ 0.43, against 0.14 and 0.05 for the paper's two commercial models on the same segments (Figure 8b).

4.3 DINOv3 classifier trained on ThaiRAP

Our second image model also puts one linear head per attribute on a frozen backbone, but the backbone is DINOv3 ViT-L/16 (Siméoni et al. 2025) and each image's features are pooled five ways: the global token, the mean patch, and the left, centre and right thirds of the image. The side pools matter because iRAP codes many attributes separately for the driver and passenger side; Thailand drives on the left, so the passenger side is the image's left. Each training image is also used mirrored, with its driver- and passenger-side labels swapped, which doubles the data for side attributes. The heads cover 30 ThaiRAP attributes with 109 classes, against the 11 read by the CLIP probes, and are trained on V-RoAst's ThaiRAP labels.

For testing we held out every fifth 2 km block of the survey route, so neighbouring images never fall on both sides of the split: 1,573 training images and 464 test images on 119 segments. Over the 25 attributes that vary on those segments, mean κ is 0.34, against 0.11 and 0.04 for the paper's Gemini-1.5-flash and GPT-4o-mini outputs on the same segments (Figure 8a). Of these, 16 pass the same gate as the probes: motorcycle flow, area type, median type, roadside object (both sides), roadside distance (passenger side), paved shoulder (both sides), property access, number of lanes, curvature, skid resistance, pedestrian crossings, traffic calming, sidewalk (passenger side) and service road.

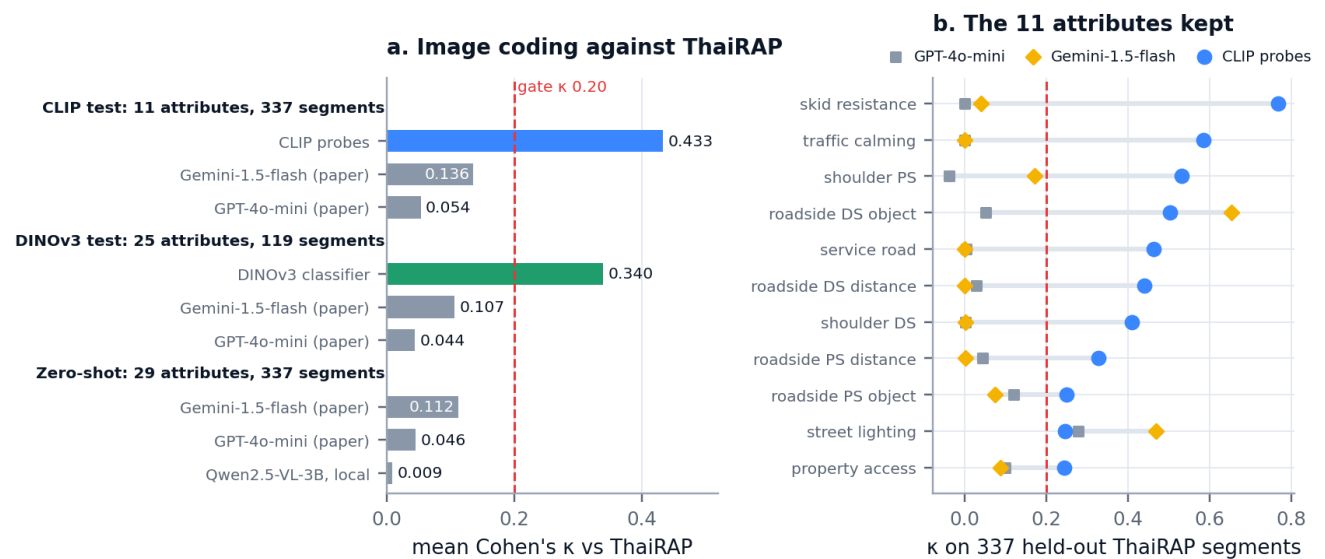


Figure 8. a: mean Cohen's κ against ThaiRAP labels; each group compares models on the same attributes and segments. b: per attribute, the CLIP probes against the paper's VLM outputs (PS/DS = passenger/driver side).

The heads are run on every usable Thai street image, and each segment gets the mean class probability per attribute. Only attributes with a design role may enter a speed model. Those that match an input of the safe-speed rules stay out (Section 3): who uses the road, its junctions and its median, such as area type,

sidewalks and median type (median separation decides the vision motorway rung). Lane count read from the images enters, as in the vision model; land use enters too, and only this member reads it. The 70 class probabilities of its 19 design-role attributes feed an optional eighth member of the research ensemble (—stack with_dinov3; Section 4.4). All 19 enter, including 8 that fail the κ gate (11 pass); the trees decide how much to use them.

4.4 Research ensemble

The research ensemble tests how well speed can be predicted with every signal we have, on the 46,548 road pieces with at least 30 probe passes. Rows are weighted by reliability, $w = n/(n + 200)$, and the five folds are 50 km blocks with each named street and carriageway pair kept whole. The score never uses it.

Gradient-boosted trees read 15 context features, or 28 with 13 city2graph measures of the road network. Two GraphSAGE networks (Hamilton et al. 2017) also read their neighbours' features, one on the graph of shared endpoints and one on a 12-nearest-neighbour graph (Figure 10). Image members add the vision model's 43 street-image columns to the city2graph tree model and to both networks, so the default stack has seven members: three tree models and four graph networks. The DINOv3 member adds the classifier's 70 class probabilities to the image tree member (141 features). A non-negative least-squares stack, fitted within each fold, combines the members.

Without images the four context members reach R^2 0.665 (Maharashtra 0.491, Thailand 0.641; Figure 9a). city2graph adds little: +0.014 R^2 over the base trees (Figure 9a), and +0.010 in an unweighted ablation, less than latitude and longitude (+0.024; Figure 9b).

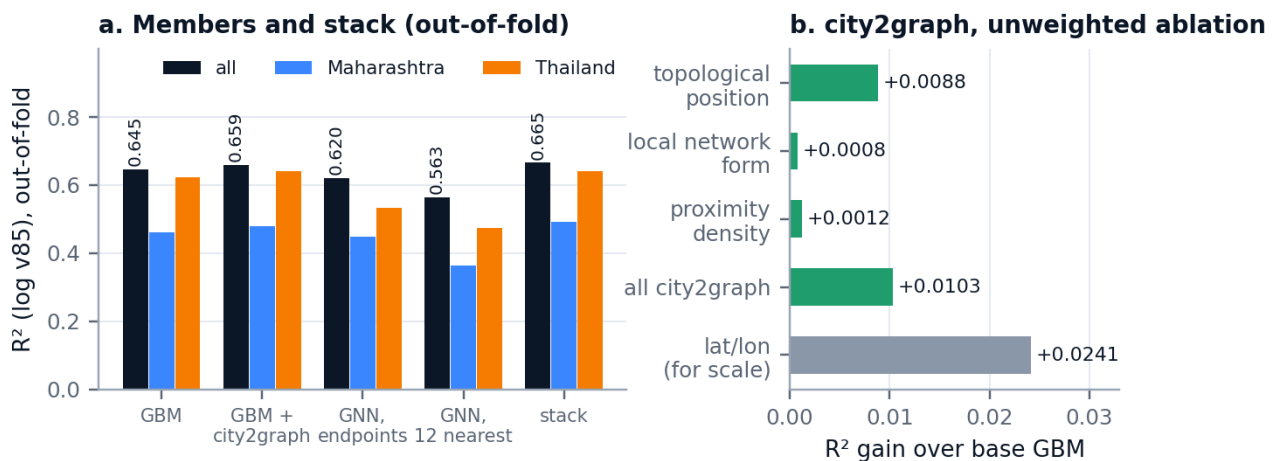


Figure 9. a: out-of-fold R^2 of the context members and their stack, overall and by country. b: gain of each group of city2graph features over the base trees, with coordinates for scale.

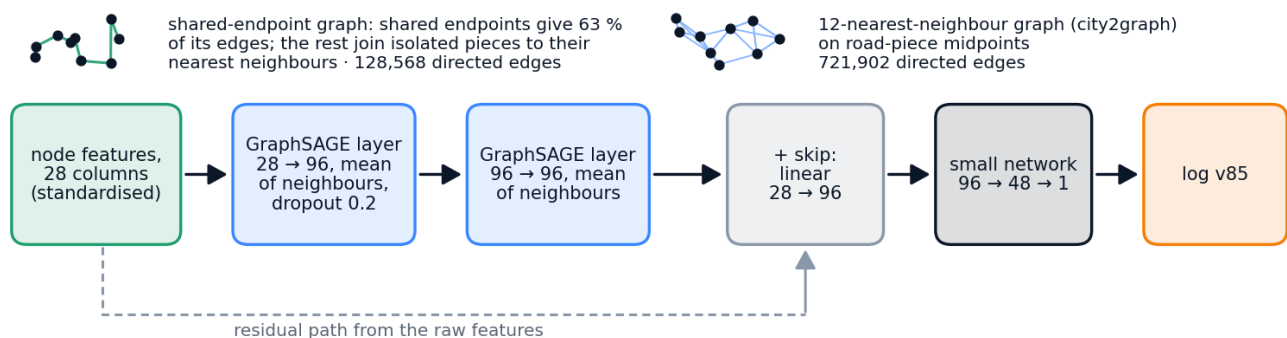


Figure 10. The GraphSAGE member: two layers that average each road piece's neighbours, a skip path from the raw features and a small network on top. The same network runs on both graphs. Road pieces with no shared endpoint are joined to their nearest neighbours.

Images help where they exist. On the Thai road pieces with imagery, the CLIP image members lift the stack from 0.638 to 0.672 (+0.034, 95 % block-bootstrap interval +0.022 to +0.044). Shuffling the image values among segments removes the gain (0.637). The gain sits in the tree members. The graph networks are unstable: storing the same image inputs at a different numeric precision changes their R^2 by up to 0.024, so we do not credit their changes to the images (Figure 11). Outside the imagery the stacks score the same (+0.004 on Thai road pieces without images, −0.001 in Maharashtra).

The DINOv3 member adds little on top. On the same Thai rows it moves the stack from 0.672 to 0.673 (+0.0010, interval +0.0003 to +0.0017); the tree member with DINOv3 scores is no better than the same member without them (difference +0.0000, interval −0.0066 to +0.0050). With the scores shuffled among segments the stack reaches 0.674, as high or higher, so none of the gain is credited to what the classifier reads. The package runs the seven-member stack by default; `--stack with_dinov3` adds the DINOv3 member and `--stack without_images` keeps the four context members (Table 2). The package's members are refitted on all 46,548 road pieces, so their predictions on those pieces are in-sample; the R^2 values in Table 2 are out-of-fold.

Stack	Members	All	MH	TH	TH with images	TH without images
without_images	4	0.665	0.491	0.641	0.638	0.640
default	7	0.669	0.490	0.661	0.672	0.644
with_dinov3	8	0.669	0.489	0.661	0.673	0.644

Table 2. Out-of-fold R^2 (log v85) of the ensemble's stacks, named as in the package, which runs default unless given `--stack`.

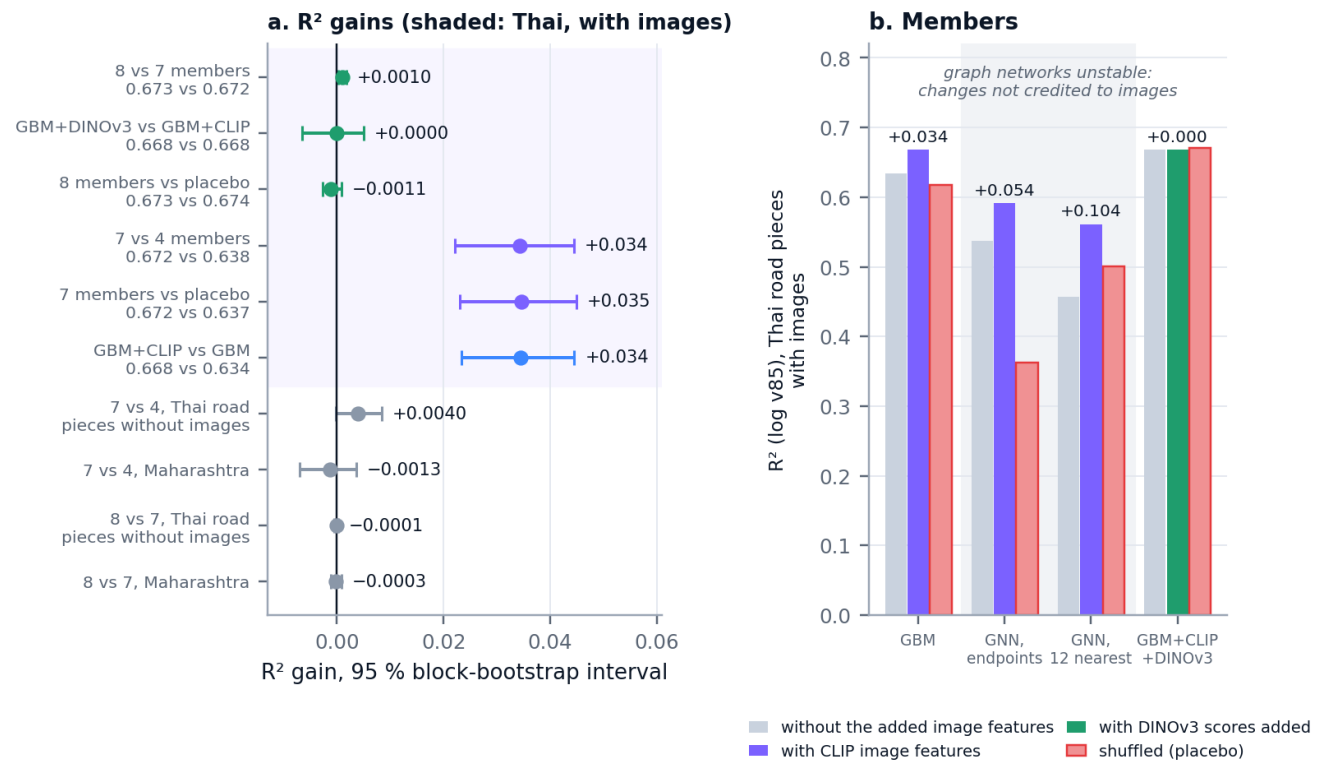


Figure 11. a: R^2 gains with 95 % block-bootstrap intervals; shaded rows are Thai road pieces with imagery. b: members on the same rows without the added image features, with them, and with them shuffled among segments (placebo); in the last group the grey bar is GBM+CLIP and the green bar adds the DINOv3 scores.

5. Safe speed and expected harm

The injury-risk curves come from Lubbe, Wu & Jeppsson (2022), fitted to German in-depth crash data for pedestrians, cyclists, motorcyclists and car occupants struck by the front of a car (Figure 12). We use killed or seriously injured at age 45. In a head-on crash the closing speed is twice the travel speed. The motorcyclist curve rests on 424 cases, 12 of them fatal, and is the least certain.

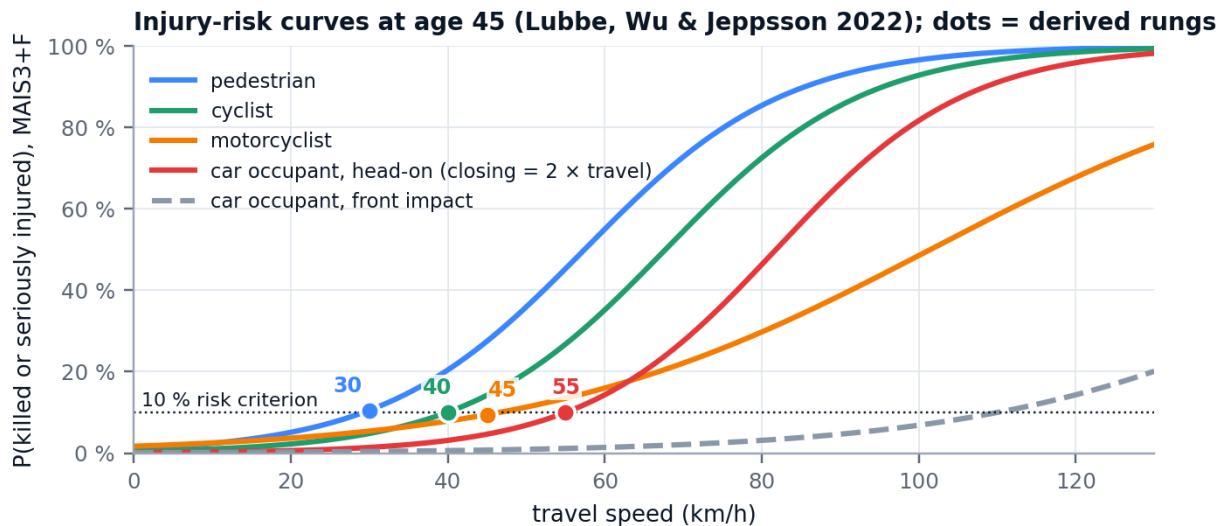


Figure 12. Risk of death or serious injury by travel speed. Dots mark where each curve reaches 10 %, rounded to 5 km/h. The side-impact and motorway rungs keep policy values.

Each rung of the safe-speed ladder is the travel speed at which its curve reaches 10 % risk: pedestrian 30 km/h, cyclist 40 km/h, motorcyclist 45 km/h, head-on 55 km/h. The policy value often quoted for head-on crashes, 70 km/h, carries 27.5 % risk on this curve. Side impact (50 km/h) and motorway (100 km/h) fall outside the source's crash scenario and keep their policy values (Figure 13).

A segment gets the rung of its most vulnerable likely conflict. GHS-SMOD ranks each 1 km cell from 0, the most rural, to 6, an urban centre. In dense settlement (classes 5–6: urban centres and dense urban clusters) people on foot mix with traffic, so the pedestrian rung applies; semi-dense settlement (classes 3–4: semi-dense urban clusters and suburban or peri-urban areas) gets it too unless the road has three or more lanes. Where OpenStreetMap has no lane tag, the lane count is predicted by gradient-boosted trees trained on the tagged roads from TomTom's road class, length, geometry, population, nearby POIs, settlement class and location. This lane model is part of the safe-speed rule, not a speed model, so it may read settlement class. In rural settlement (classes 0–2) we assume motorcycles share the road. Motorways get the motorway rung, and segments without settlement data fall back to road class and land use. The pedestrian rung applies to 5,771 segments, the motorcyclist rung to 9,601 and the motorway rung to 182.

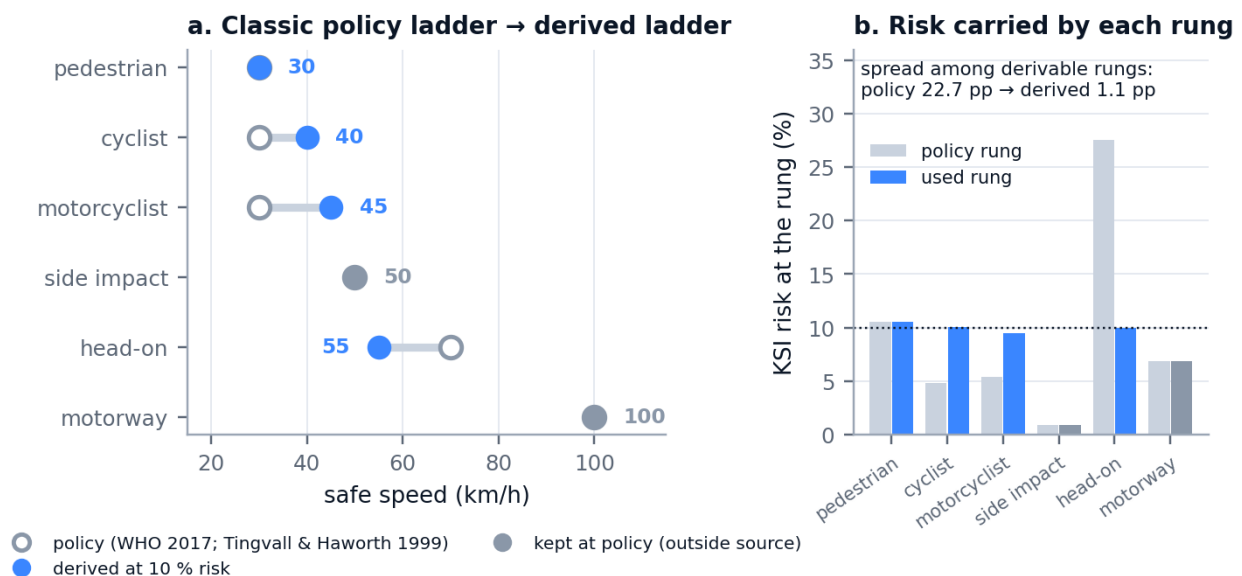


Figure 13. a: the policy ladder (WHO 2017; Tingvall & Haworth 1999) and the ladder derived from the curves. b: the injury risk each rung carries.

Expected harm is the injury risk of the segment's crash type averaged over its speed distribution. Where TomTom profiles cover the road, harm is measured; elsewhere it comes from the design-speed model's v50 and v85 through the same curve. The two are blended by $w = n/(n + 200)$ on probe passes (Figure 14). The median segment has 169,393 passes, so measurement dominates almost everywhere. Harm rebuilt from v50 and v85 alone matches harm from the full profile with R^2 0.946.

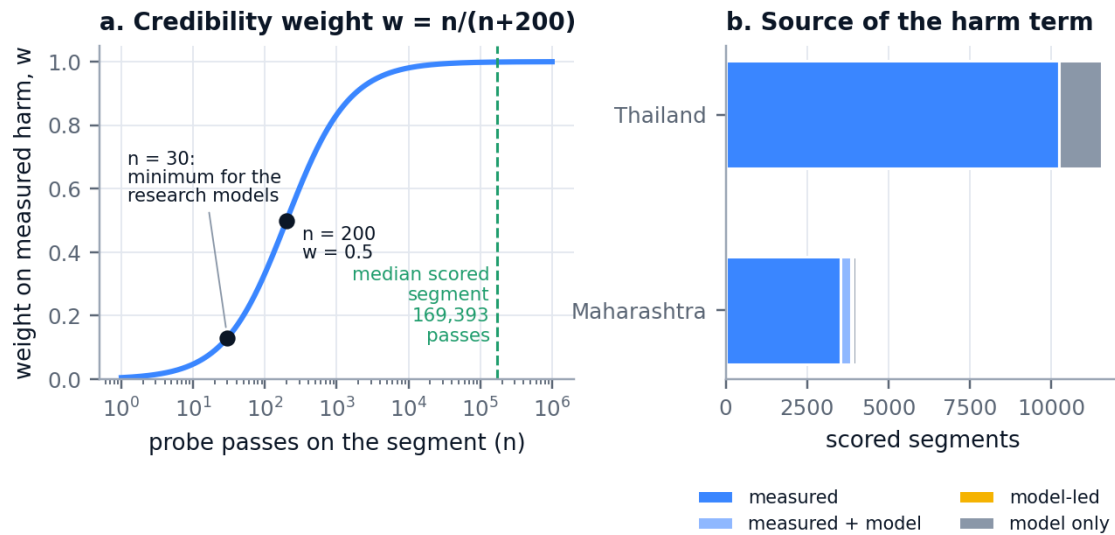


Figure 14. a: weight on measured harm by probe passes. b: where the harm term comes from, by country: measured ($w \geq 0.75$), measured + model ($0.25 \leq w < 0.75$), model-led ($w < 0.25$) and model only (no TomTom profile).

6. Score and recommended action

Each segment's score is a weighted sum: $\text{score} = 100 \times (0.35 \cdot \text{mismatch} + 0.30 \cdot \text{harm} + 0.20 \cdot \text{vru} + 0.15 \cdot \text{volume})$. Each part is scaled to 0–1, and the points are rounded so that they add up to the score exactly. Harm enters as a rank within the country, so it orders roads inside a country but does not compare the two countries (Table 3, Table 4).

Part	Definition	Weight
mismatch	(posted limit – safe speed) / 45 km/h, clipped to 0–1	0.35
harm	within-country percentile of the blended expected harm	0.30
vru	$0.50 \times \text{heuristic} + 0.30 \times \text{POIs} + 0.20 \times \text{population}$. The heuristic is 0.70 on urban roads or 0.30 on rural ones, plus a road-class adjustment (secondary +0.10, primary +0.05, trunk –0.05, motorway –0.55), plus $0.15 \times$ the share of riders without a helmet, clipped to 0–1. POIs (a count weighted by relevance to people on foot: schools and transport stops 1.0, restaurants 0.7, hospitals 0.5, fuel stations 0.3, police and fire stations 0.2) and population are put on a log scale and scaled between the 5th and 95th percentile	0.20
volume	log probe count, scaled between the 5th and 95th percentile	0.15

Table 3. The four parts of the score.

Band	Score	Meaning: how soon to act
Aligned	0–24	Lowest urgency — no action beyond the routine review.
Low	25–44	Low urgency — monitor.
Moderate	45–59	Review the limit at the next routine review.
High	60–77	Prioritise for limit review and engineering.
Critical	78–100	Most urgent — start here.

Table 4. Bands.

Each segment gets one recommended action. Where the limit exceeds the safe speed by more than 5 km/h (14,791 segments), the first option is to lower it to the safe speed. Its effect on fatal and serious injuries (FSI) follows the Power Model (Nilsson 2004; Elvik et al. 2019) on the segment's own speeds: $\text{FSI change} = (v_{\text{after}} / v_{\text{before}})^{3.0} - 1$. We report a realistic case, in which 60 % of the change reaches operating speeds and the effect is capped at 50 %, and a fully enforced case; their median FSI reductions are 50 % and 88 %. Being nearly free, the limit change wins the cost-effectiveness ranking wherever it is offered.

Engineering options come from 14 iRAP Toolkit countermeasures with their published reduction ranges, cost bands and service lives. The best one is reported beside the limit change, because a limit works only where the road supports it. A treatment whose premise is a missing facility is dropped where street images already show that facility (5,137 segments). Actions are ranked by harm reduced per US dollar per km per year.

7. Verification

We check the method against published evidence, independent labels and its own arithmetic (Table 5). The strongest external check is the World Bank's Thailand speed study (2019), which published iRAP-modelled FSI changes for 8 limit changes on two roads. Our Power Model reproduces all of them within 2.5 pp (Figure 15a).

Check	Result	Status
Published evidence and the safe-speed ladder		
World Bank (2019) Thailand case study	8 published FSI changes reproduced; worst gap 2.5 pp	pass
Injury curves vs Lubbe et al. (2022)	worst gap 1.37 pp over 16 published risks (Figure 15b)	pass
Pedestrian curve vs Rosén & Sander (2009)	largest gap 5.7 pp at 30/50/70 km/h	pass
Policy ladder (WHO 2017) vs derived rungs	risk spread across rungs 22.7 pp (policy) → 1.1 pp (derived)	pass
Image attributes against ThaiRAP		
Our scoring of V-RoAst's outputs vs the V-RoAst paper's Table 1	Gemini accuracy 0.818, F1 0.471; GPT-4o-mini 0.751, 0.418, as published	pass
Zero-shot VLM coding	mean κ 0.11 (Gemini), 0.05 (GPT-4o-mini), 0.01 (Qwen2.5-VL-3B)	fail: not used
CLIP probes, same attributes as the VLMs	κ 0.43 vs 0.14 (Gemini), 0.05 (GPT-4o-mini)	pass
DINOv3 classifier, same segments as the VLMs	κ 0.34 vs 0.11 (Gemini), 0.04 (GPT-4o-mini)	pass
Probes on Mapillary images	2 of 5 ThaiRAP-probe outputs reach $\rho \geq 0.20$ with Mapillary detections (the OSM divided-road probe also fails) (Figure 16)	weak
Models and score		
Exact attribution	largest residual 6×10^{-14} ; score points add up exactly	pass
Ensemble reproduction	4 context members retrained from scratch reproduce the reported out-of-fold predictions exactly	pass
Image gain without segments sharing images across folds	+0.033 (interval +0.022 to +0.044), 78 segments dropped	pass
Image gain vs placebo	real over shuffled images +0.035 (interval +0.023 to +0.045)	pass
DINOv3 member vs placebo	real over shuffled scores -0.0011 (interval -0.0027 to +0.0008); the package runs the seven-member stack by default and offers --stack-with_dinov3	fail: no real gain
Ranking stability	$\rho \geq 0.986$ with the limit-gap weight or all rungs $\pm 20\%$ (Figure 17)	pass
Vulnerable road users: method vs street images	whether the method expects people on foot or two-wheelers in traffic, against whether Mapillary detections see them: κ 0.16 on 8,065 segments	weak
Transfer between countries	R^2 -0.321 (MH→TH), -0.123 (TH→MH)	fail: not claimed

Table 5. Verification checks, including the weak results and failures.

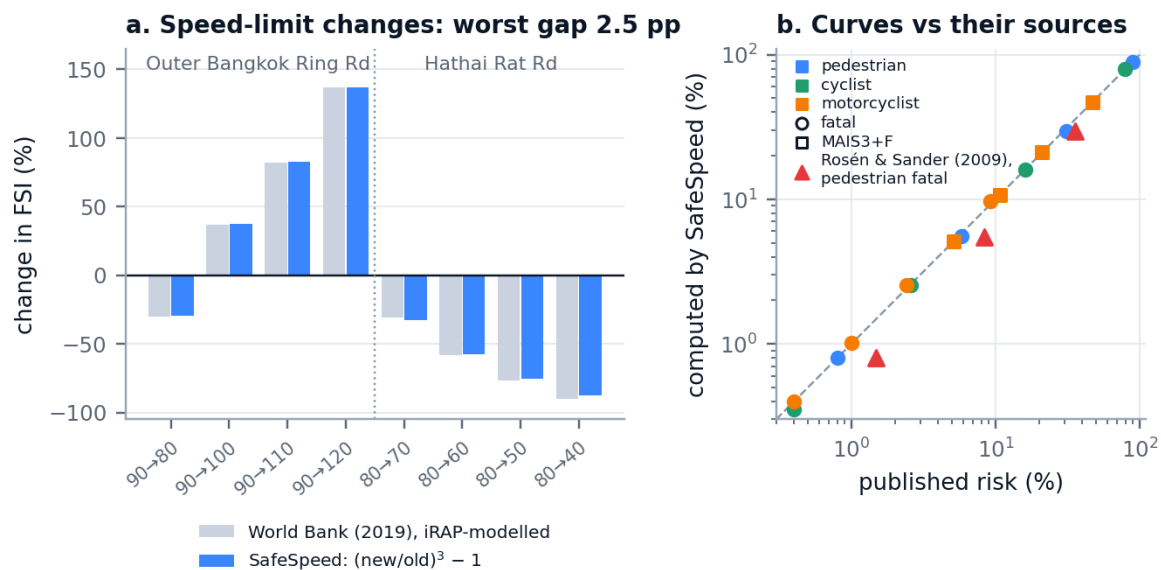


Figure 15. a: FSI change for the World Bank's limit changes, as published and as computed with exponent 3.0. b: published injury risks against the same risks from our curves.

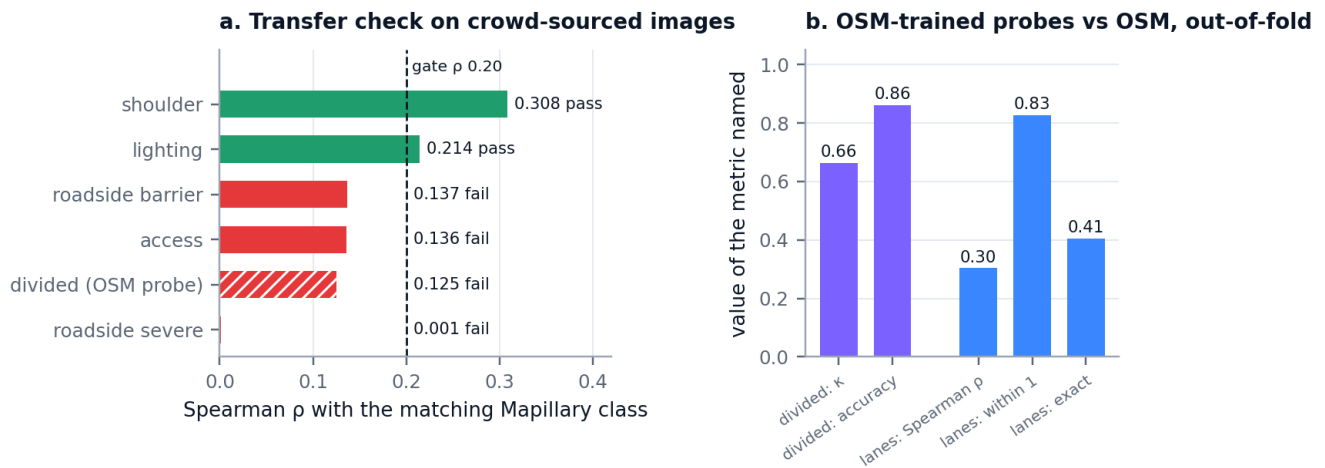


Figure 16. a: ThaiRAP-trained probes (solid) and the OSM-trained divided-road probe (hatched) against the matching Mapillary detections. b: the OSM-trained probes against OSM tags.

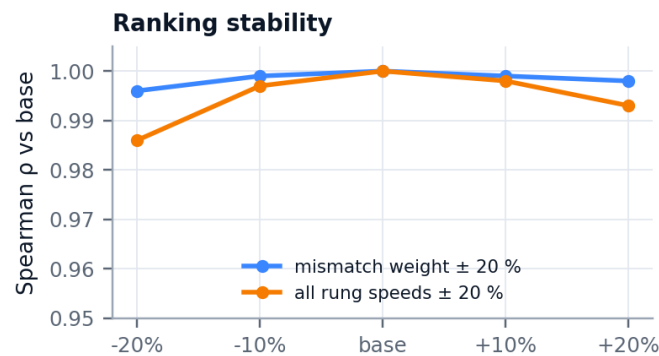


Figure 17. Rank correlation with the base ranking when the limit-gap weight or all rung speeds change by up to 20 %.

8. Results

Table 6 collects the main numbers. The score is not a speeding index: its rank correlation with the share of vehicles over the limit is 0.039, and the top 1,000 roads of the two lists barely overlap (Figure 18).

Result	Value
Segments scored	15,554 (Maharashtra 4,010, Thailand 11,544)
Road length in the High or Critical band	Maharashtra 16 %, Thailand 72 %
Segments whose harm uses a TomTom profile	91 % of segments
Design-speed model, R^2 log v85	0.607 pooled; 0.416 MH; 0.512 TH
Vision model (Thailand), R^2 log v85	0.437 (context on the same rows 0.523)
Research ensemble, R^2 log v85	0.669 default stack; 0.665 without images; all road pieces
Research ensemble, Thai road pieces with images	0.638 without images; 0.672 with CLIP features (the package default); 0.673 with DINOv3 too
World Bank Thailand check, worst gap	2.5 pp

Table 6. Key results.

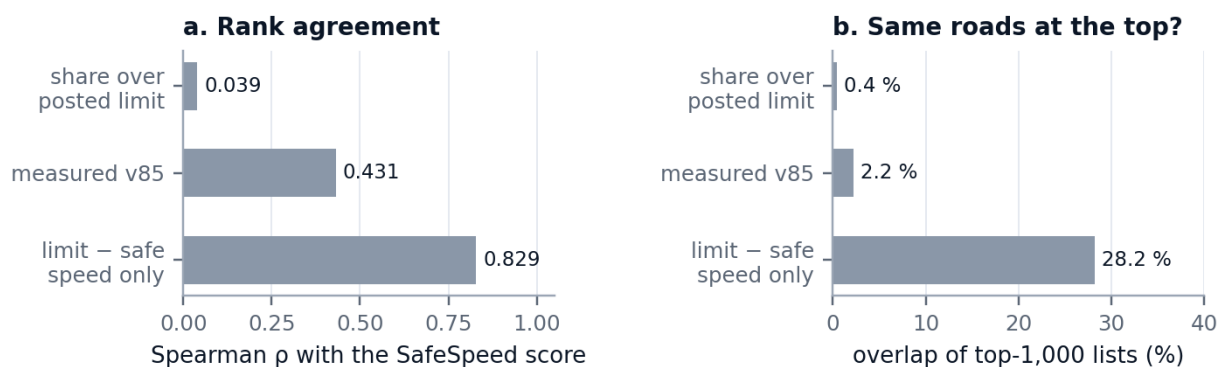


Figure 18. Rank agreement and top-1,000 overlap between the SafeSpeed score and simpler rankings.

A High or Critical band covers 72 % of the analysed road length in Thailand and 16 % in Maharashtra (Figure 19). Posted limits exceed the safe speed on 14,916 segments, and the limit gap contributes most to the score in Thailand and expected harm in Maharashtra (Figure 20).

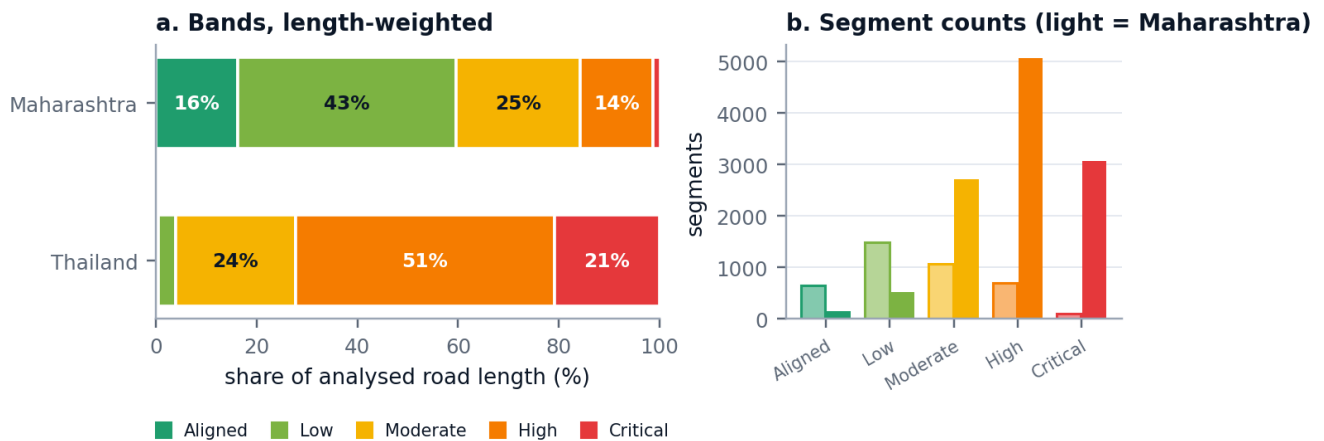


Figure 19. Bands by country. a: share of analysed road length. b: number of segments.

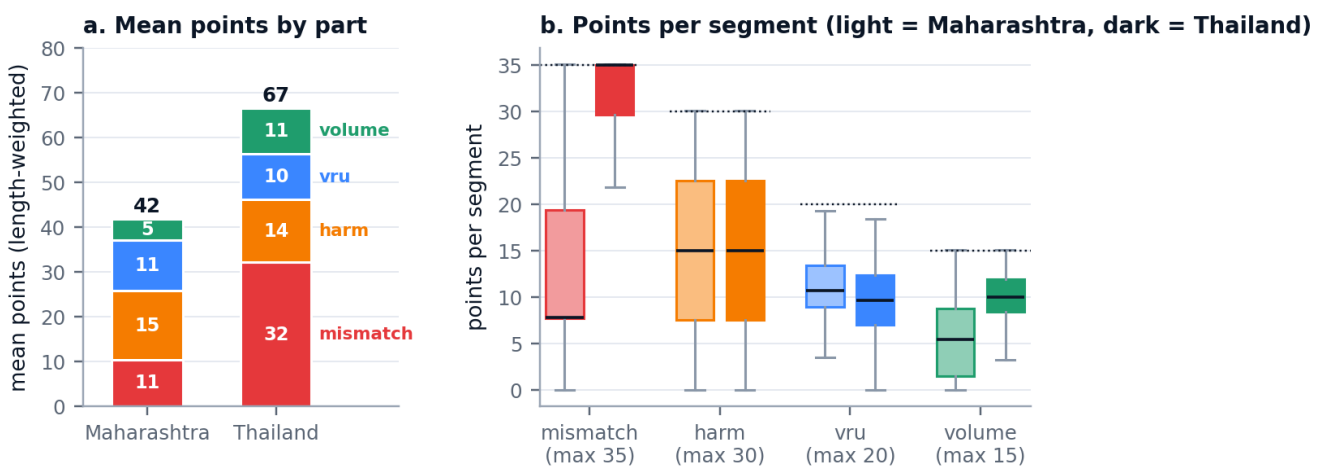


Figure 20. What drives the score. a: mean points per part, length-weighted. b: points per segment; dotted lines mark each part's maximum.

Katraj - Dehu Road Bypass (MH-3983, Maharashtra) shows the arithmetic. It is posted at 80 km/h in dense settlement, so its safe speed is 30 km/h (pedestrian impact) and it scores 96 (Figure 21).

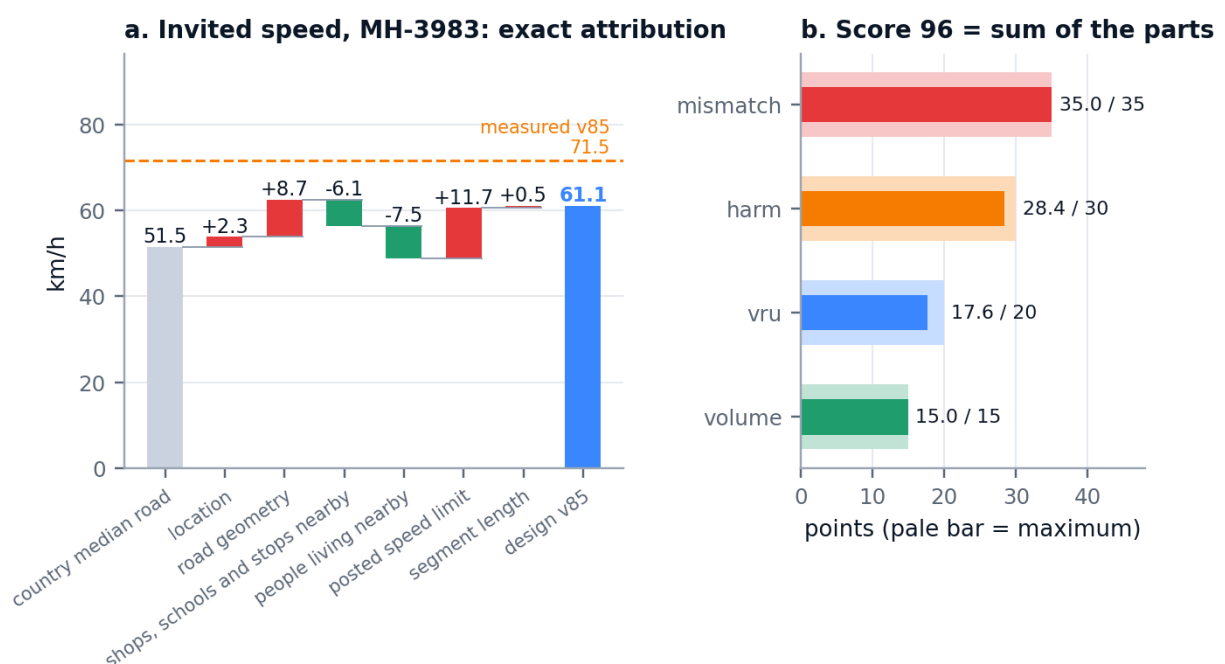


Figure 21. Worked example, segment MH-3983. a: the country's median road plus each factor group's contribution gives the design speed exactly. b: the score is the sum of its four parts.

9. Limitations and licences

There are no crash records, so nothing here is validated against casualties. The outputs are conditional on a crash happening, and the cost-effectiveness figure ranks actions but is not a benefit-cost ratio.

The injury curves come from German data on impacts with the front of a car, and travel speed is not impact speed. The side-impact and motorway rungs are policy values. TomTom probes are mostly cars and trucks, with no motorcycle class, and we received a single time period, so free flow and congestion are mixed.

Many posted limits are modelled rather than surveyed, and 843 were missing and imputed; the limit gap carries the largest weight. Lane counts are mostly imputed, and the semi-dense rung depends on them. Probe counts reflect fleet share as well as traffic. On a small set of well-measured roads the measured part of the harm blend takes its crash type from road class and land use, which can differ from the crash type behind the safe speed; this moves harm only slightly.

The speed models do not transfer between countries. The graph networks add little and react to small input changes. The image models cover Thailand only, the ThaiRAP-trained probes transfer only partly to Mapillary images, and the external agreement checks are weak (agreement with street-image detections of vulnerable road users κ 0.16; vision against headline safe speed κ 0.15).

The ThaiRAP survey images are licensed CC BY-NC 4.0, and the ThaiRAP label table, taken from the V-RoAst repository, carries no declared licence. Everything trained on them is distributed for non-commercial use only under CC BY-NC 4.0: the CLIP probes, the DINOv3 heads, the vision model, the image members of the ensemble and their outputs. The DINOv3 backbone is not included; it is downloaded under Meta's DINOv3 License, and a copy of that licence accompanies the DINOv3 classifier heads and the cached DINOv3 features of the sample images, which are distributed under it. Mapillary images and detections are CC BY-SA 4.0, OpenStreetMap data ODbL and WorldPop CC BY 4.0. The TomTom speeds, posted limits and feature files are used under the terms of the AI for Safer Roads 2026 Innovation Challenge.

Appendix A. Variables

'ref-only' marks the columns that build the safe speed or describe who uses the road; they never enter a speed model.

TomTom's road class (frc), the lane count read from images and, in the optional DINOv3 member, land use read from images do enter (Section 3).

Variable	Meaning	Source	Used by
Challenge road network and TomTom aggregate speeds			
road_class, land_use	motorway to secondary; urban or rural (GRUMP urban extents; CIESIN et al. 2011)	RoadClass, LandUse	safe-speed fallback, VRU heuristic; ref-only
posted_limit, posted_used	posted limit, km/h; 0 → missing → (country, class) median	SpeedLimit	limit gap; model feature
median_speed, operating_speed	TomTom v50 and v85, km/h	MedianSpeed, F85thPercentileSpeed	model targets
pct_over_limit	share of vehicles over the limit	PercentOverLimit	comparison only
volume_proxy	probe count	Sample_Size_Total	volume part
length_km, lon, lat	length; first vertex	RoadLength; geometry	features, folds
helmet_rate	helmet wearing by region and land use	helmet-use survey, by region and urban or rural (Asian Development Bank 2025)	VRU heuristic; ref-only
Challenge feature files, joined 1:1 on the segment key			
poi_school ... poi_safety	counts of schools, hospitals, restaurants, stops, fuel and safety sites	*Count fields	VRU part; model features
poi_vru_raw	POI count weighted by relevance to people on foot: schools and transport stops 1.0, restaurants 0.7, hospitals 0.5, fuel stations 0.3, police and fire stations 0.2	*Count fields	VRU part; model feature
seg_helmet	observed helmet use on the segment	OverallHelmetUse	reporting only; ref-only
Population, settlement, OpenStreetMap			
pop_exposure, pop_density	residents within 300 m; mean per 100 m cell	WorldPop 2025	VRU part; features
smod_rank, smod_degurba	settlement class, 0 (most rural) to 6 (urban centre); degree of urbanisation	GHS-SMOD R2023A	safe speed; ref-only
osm_lanes, lanes_imputed, oneway	lane count (imputed where untagged); one-way share	OSM tags; untagged roads predicted by gradient-boosted trees	safe speed (lanes); probe labels; ref-only

Variable	Meaning	Source	Used by
Road geometry and graph			
geo_sinusosity, geo_curvature	length ÷ straight-line distance; turning per metre	linework	model features
geo_junction_density	other segment ends within 35 m, per km	network	model feature
c2g_*	graph measures: degree, PageRank, betweenness, neighbourhood size, local network form	city2graph	research ensemble
TomTom Traffic Stats, per road piece			
P5 ... P95	19 speed percentiles, km/h	speedPercentiles	speed distribution
frc	TomTom functional road class 0–7 (0 = motorway)	TomTom Traffic Stats	research ensemble
sample_size, reliability	probe passes; $n/(n + 200)$	sampleSize	weights, harm credibility
harm_measured	$\int R(v) dF(v)$ for the segment's crash type	percentiles × curve	harm blend
Street images, Thailand			
v_shoulder, v_roadside_*, v_skid, v_lighting, v_calming, v_access, v_service_road	ThaiRAP attributes read by the CLIP probes	CLIP ViT-L/14 + probes	vision model; ensemble
v_lanes, v_divided	lane count; divided road	CLIP + OSM-trained probes	vision model; divided: vision safe speed only
v_ped_seen, v_moto_seen, v_sidewalk, v_crossing, ...	share of images with people, riders, footpaths, crossings	Mapillary detections	vision safe speed; ref-only
v_emb_00 ... v_emb_31, has_vision	principal components of the mean CLIP embedding; 1 if images exist	PCA	vision model; ensemble
a_<attribute>__<code>	mean probability of each iRAP code over the segment's images	DINOv3 classifier	ensemble, with_dinov3 stack only (all 19 design-role attributes, gated or not)
mly_*_imgs	images showing crossings, footpaths, lights, signals, people	Mapillary API	action checks, the street-image check (Table 5); never a feature
Outputs per segment			
safe_speed, safe_speed_crash_type_key, harm, harm_source	safe speed and its crash type; blended harm and where it comes from	ladder rule; design-speed model + TomTom	limit gap, harm part
design_speed_v50/v85, speed_attr_*, pts_*, score, band	invited speed and its split by factor group; points per part, score, band	design-speed model; score	priority
vis_*; action, action_engineering, limit_reduction_fsi_pct	vision-only speed, harm and score; recommended action, best engineering option, FSI change of a limit cut	vision model; iRAP Toolkit; Power Model	beside the score; action

References

- Asian Development Bank (2025). Road Safety Performance Indicators: helmet-wearing rates for Thailand and Maharashtra, by region and urban or rural.
- Asian Development Bank & Agilysis (2026). AI for Safer Roads 2026 Innovation Challenge dataset: road networks of Maharashtra and Thailand with TomTom speeds, posted limits and feature files.
- Bai, S. et al. (2025). Qwen2.5-VL technical report. arXiv:2502.13923.
- CIESIN, IFPRI, World Bank & CIAT (2011). Global Rural-Urban Mapping Project, Version 1 (GRUMPv1): Urban Extents Grid. NASA Socioeconomic Data and Applications Center, Palisades, NY. doi:10.7927/H4GH9FVG.
- Elvik, R., Vadeby, A., Hels, T. & van Schagen, I. (2019). Updated estimates of the relationship between speed and road safety at the aggregate and individual levels. *Accident Analysis & Prevention* 123:114–122.
- European Commission JRC (2023). GHS-SMOD R2023A. doi:10.2905/A0DF7A6F-49DE-46EA-9BDE-563437A6E2BA.
- Hamilton, W., Ying, R. & Leskovec, J. (2017). Inductive representation learning on large graphs. *Advances in Neural Information Processing Systems* 30.
- iRAP (2024). Road Safety Toolkit. toolkit.irap.org.
- Jongwiriyanurak, N. et al. (2025). V-RoAst: Visual Road Assessment. ICCV Workshops. doi:10.1109/ICCVW69036.2025.00176. ThaiRAP survey images: UCL Research Data Repository, doi:10.5522/04/26520787.v1, CC BY-NC 4.0; label table: github.com/PongNJ/V-RoAst, no declared licence.
- Lubbe, N., Wu, Y. & Jeppsson, H. (2022). Safe speeds: fatality and injury risks of pedestrians, cyclists, motorcyclists, and car drivers impacting the front of another passenger car as a function of closing speed and age. *Traffic Safety Research* 2:000006. doi:10.55329/vfma7555.
- Mapillary (2026). Street-level imagery and object detections. mapillary.com, CC BY-SA 4.0.
- Meta (2025). DINOv3 License. ai.meta.com/resources/models-and-libraries/dinov3-license.
- Nilsson, G. (2004). Traffic safety dimensions and the Power Model to describe the effect of speed on safety. *Bulletin* 221, Lund Institute of Technology.
- OECD/ITF (2018). Speed and Crash Risk. International Transport Forum, Paris.
- OpenStreetMap contributors (2026). OpenStreetMap. openstreetmap.org, ODbL 1.0.

- Radford, A. et al. (2021). Learning transferable visual models from natural language supervision. Proceedings of ICML, PMLR 139.
- Rosén, E. & Sander, U. (2009). Pedestrian fatality risk as a function of car impact speed. Accident Analysis & Prevention 41(3):536–542.
- Sato, Y. (2026). city2graph, version 0.4.0: graph representations of urban data for geospatial deep learning. github.com/c2g-dev/city2graph, BSD-3-Clause.
- Siméoni, O. et al. (2025). DINOv3. arXiv:2508.10104.
- Tingvall, C. & Haworth, N. (1999). Vision Zero: an ethical approach to safety and mobility. 6th ITE International Conference, Melbourne.
- TomTom (2026). Traffic Stats: speed percentiles and probe sample sizes per road piece, provided through the AI for Safer Roads 2026 Innovation Challenge.
- WHO (2017). Managing Speed. World Health Organization, Geneva.
- WHO (2023). Global status report on road safety 2023: country profiles for India and Thailand. World Health Organization, Geneva.
- World Bank (2019). Speed Variation Analysis: A Case Study for Thailand's Streets and Roads. World Bank, Washington, DC.
- WorldPop (2025). Constrained population counts, 100 m, R2025A. CC BY 4.0.